

Appendix: PHIRL: Aligning Learned Rewards with Task Progress for Inverse Reinforcement Learning

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A. IMPLEMENTATION DETAILS

A. Additional Details for Demonstration Annotation

While all demonstrations will be used for reward inferring, PHIRL only requires a part of the demonstrations to be annotated. **AIRL** purely learns from human demonstrations. It learns to distinguish expert trajectories from policy roll-outs while being regularised to learn a reward function that is disentangled from transition dynamics. **Pref** learns solely from human preference feedback. It learns a reward model from pairwise comparisons of trajectory clips. **AIRLPref** is adapted from DemoPref by replacing the reward learner with AIRL to avoid manually designing the reward features. DemoPref first fits a Gaussian-process-based reward function from demonstrations, uses that prior to train a generator to generate preference queries, and further updates the learned reward function. We trained all baselines and PHIRL for 500 episodes, with 10 parallel environments and an environmental horizon of 256 for Lift and PickPlaceCan.

B. Hyperparameter

We use a hyperparameter search method from Optuna. The hyperparameters we tuned are:

- **Environment & network geometry**
 - Number of parallel environments, n_{envs} : {1, 5, 10}
 - Actor hidden units (`reward_size`): {32, 48, 64, 128, 256}
 - Critic hidden units (`potential_size`): {24, 32, 64, 128}
- **PPO optimisation**
 - Learning rate, η (`learning_rate`): 1×10^{-4} – 3×10^{-4}
 - Discount factor, γ : {0.70, 0.85, 0.88, 0.90, 0.92, 0.95, 0.97, 0.99}
 - GAE parameter, λ : {0.80, 0.85, 0.90, 0.92, 0.95, 0.98}
 - Entropy coefficient (`ent_coef`): 0.001–0.051 in steps of 0.005
 - PPO clipping range (`clip_range`): 0.10–0.40 in steps of 0.10
 - Time-steps per update, $n_{\text{steps}} = \text{horizon} \times f$, where f (`n_steps_factor`): {0.25, 0.50, 1.0, 2.0}
- **Generator / replay buffer**
 - Generator-buffer size multiplier (`gen_buff_multiplier`): {1, 2, 3, 4}
- **Discriminator & reward shaping**

- Number of discriminator updates per round, n_{disc} : {2, 4, 8}
- Demonstration-batch ratio (`demo_batch_size`): 0.01–0.31 in steps of 0.02
- Shaping batch size (`shaping_bs`): {4, 8, 12, 16}

C. A Second Version of PHIRL Online

We implemented a second version of PHIRL online, PHIRL FreshOnline. In PHIRL FreshOnline, the sampled trajectories only stayed in the annotation demonstration set till next round of sampling. In this case, the PHIRL always sees the freshest data from the generator and thus improves learning efficiency. We believe this method will be powerful, especially when the task is long-horizon. However, in our pilot test, we found that this method requires significantly more data to be effective, which conflicts with the design intuition of PHIRL. Thus, we did not include the method in our paper. Nevertheless, the implementation of this version can still be found in our code.

B. DATA COLLECTIONS AND VLM ATTEMPTS

The instructions participants received were: "Based on the current trajectory, how much do you think the robot has completed the task? Zero means totally incomplete, and 100 means fully completed." Each participant annotated a third of the demonstrations for each dataset. Participants watched replays of the demonstrations in simulations. By default, PHIRL assumes 20 percent of demonstrations being annotated, resulting in 400 (600 for MH datasets) *progress* labels. We use the same amount of labels and three times the number of images (one start frame, one middle frame, and one end frame) to fine-tune the VLM model. We count one demonstration as 10 feedback and one preference feedback as one evaluative feedback to normalizing human labeling effort. Thus, Pref received preference feedback for 2,000 trajectory pairs since there are 200 demonstrations in total for PH datasets, which is equal to demonstration-only methods. AIRLPref received 400 (600 for MH) pairs of preference feedback, which is the same as PHIRL-20% (i.e., the standard version of PHIRL), since prior work indicates that providing progress has a similar task load to providing scalar evaluative feedback. PHIRL online received an additional 400/600 online *progress* annotations from a finetuned VLM since this is generate by VLM not humans.

We also used a Gemma 3 12B model to provide *progress* annotations. For Lift domain, the prompts we used are:

VLM Prompt

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{
\textit{You now are a human dataannotator. Your
task is to provide feedback to a robotic task based
on the image I provided.
I will first introduce what feedback you will use
and then introduce you to the task.
Feedback:
You will be using a signal called progress.
\textit{Progress} shows whether the robot is acting
towards task completion in a scale value and can
indicate if the task has been completed.
\textit{Progress} is independent from the past
trajectory taken, i.e.,
for any given task $t$, current state $s$, any
action $a_i$ and $a_j$, and any previous state $s_i$
and $s_j$, the \textit{progress}:
$prog_t(s_i, a_i, s) = prog_t(s_j, a_j, s) \forall i, j$
You should use \textit{progress} in a range of 0 to
100, where 100 indicates the task is fully
complete, and 0 means that no progress has been
made yet.
Task: You will see a robot arm and a red block on
the table. The goal for the robot is to lift the block
from the table.
Your job is to, based on best of your ability,
evaluate how much progress the robot has made so
far. Once you finish your evaluation, double check
it with the image you received, then give me your
response. If you are not sure, **TAKE YOUR
BEST GUESS**.
}
}
```

For the PickPlaceCan domain, the prompt we used is: *You now are a human data annotator. Your task is to provide feedback to a robotic task based on the image I provided. I will first introduce what feedback you will use and then introduce you to the task. Feedback: You will be using a signal called progress. Progress shows whether the robot is acting towards task completion in a scale value and can indicate if the task has been completed. Progress is independent from the past trajectory taken, i.e., for any given task t , current state s , any action a_i and a_j , and any previous state s_i and s_j , the progress: $prog_t(s_i, a_i, s) = prog_t(s_j, a_j, s) \forall i, j$. You should use progress in a range of 0 to 100, where 100 indicates the task is fully complete, and 0 means that no progress has been made yet. Task: You will see a robot arm, a red can, and two tables. The red can is on the one of the table, and there is a semi-transparent grey can on the other table. The grey can is not real and used to indicate the goal location. The robot's goal is to pick up the can, and drop it near the grey area. Your job is to, based on best of your ability, evaluate how much progress the robot has made*

*so far. Once you finish your evaluation, double check it with the image you received, then give me your response. If you are not sure, ****TAKE YOUR BEST GUESS****.*

A. Example of Generated Trajectory and Annotations

We demonstrate an example of how generated trajectories look and their annotations in Table 1, and Table 2. The annotations are generated by a heuristic function adapted from the environmental rewards. If a VLM is used, the values may vary, but the form will stay the same.

C. ADDITIONAL RESULTS

A. Additional analysis

The error bars represent the standard deviation, and the reported rewards are smoothed by averaging over 100 evaluation rollouts executed with the best checkpoint using 100 random seeds (from 1 to 100).

In ?? we reported some statistics for the results in the paper. Here we show some additional analysis. In the Lift-PH domain, PHIRL achieved a mean environmental reward of $M = 188.12$ ($SD = 63.08$), which is significantly higher than Rank2Reward ($M = 174.51$, $t = 1.54$, $p = 0.124$), AIRL ($M = 169.62$, $t = 2.25$, $p < 0.05$) and Pref ($M = 142.55$, $t = 4.97$, $p < 0.001$), and noticeably higher than AIRLPref ($M = 178.57$, $t = 1.12$, $p = 0.26$). The advantage of PHIRL became more significant in the Lift-MH dataset, where demonstration quality is lower. PHIRL ($M = 167.16$, $SD = 46.88$) significantly outperformed AIRL ($M = 118.30$, $t = 9.12$, $p < 0.001$), AIRLPref ($M = 130.92$, $t = 6.88$, $p < 0.001$), Rank2Reward ($M = 121.08$, $t = 8.38$, $p < 0.001$) and Pref ($M = 142.55$, $t = 3.03$, $p < 0.01$).

In the more challenging PickPlaceCan domain, where the average demonstration horizon is approximately twice of Lift, PHIRL achieved greater performance advantages. For Can-PH, PHIRL achieved $M = 152.29$ ($SD = 64.35$), significantly higher than AIRL ($M = 41.55$, $t = 14.78$, $p < 0.001$), AIRLPref ($M = 59.72$, $t = 12.47$, $p < 0.001$), Rank2Reward ($M = 79.64$, $t = 9.84$, $p < 0.001$), and Pref ($M = 4.01$, $t = 22.83$, $p < 0.001$). For Can-MH, PHIRL maintained robust performance ($M = 116.99$, $SD = 20.39$), and obtained significantly higher rewards than AIRL ($M = 18.02$, $t = 40.13$, $p < 0.001$), AIRLPref ($M = 35.71$, $t = 17.38$, $p < 0.001$), Rank2Reward ($M = 8.00$, $t = 52.01$, $p < 0.001$), and Pref ($M = 4.01$, $t = 52.87$, $p < 0.001$).

B. Reward Curves

We also include the success rates and learning curves for baselines here.

C. additional domain

We selected an additional domain for Robosuite [1]: Square. The annotation and feedback were from two participants instead of six. Other than that, the setups are the same as in Section 4. The goal is to control a robotic arm to pick up a square-shaped nut and place it onto a peg, shown

TABLE I
LIFT AND CORRESPONDING PROGRESS

Segment	Start Step	End Step	Start Progress	End Progress	Interval
0	0	26	0.0	17	26
1	26	51	17	29	25
2	51	77	29	36	26
3	77	103	36	35	26
4	103	128	35	36	25
5	128	154	36	36	26
6	154	180	36	36	26
7	180	205	36	36	25
8	205	231	36	37	26
9	231	256	37	38	25

TABLE II
CAN AND CORRESPONDING PROGRESS

Segment	Start Step	End Step	Start Progress	End Progress	Interval
0	0	51	0.0	4	51
1	51	103	4	15	52
2	103	154	15	17	51
3	154	205	17	17	51
4	205	257	17	17	52
5	257	308	17	17	51
6	308	359	17	17	51
7	359	411	17	18	52
8	411	462	18	27	51
9	462	512	27	29	50

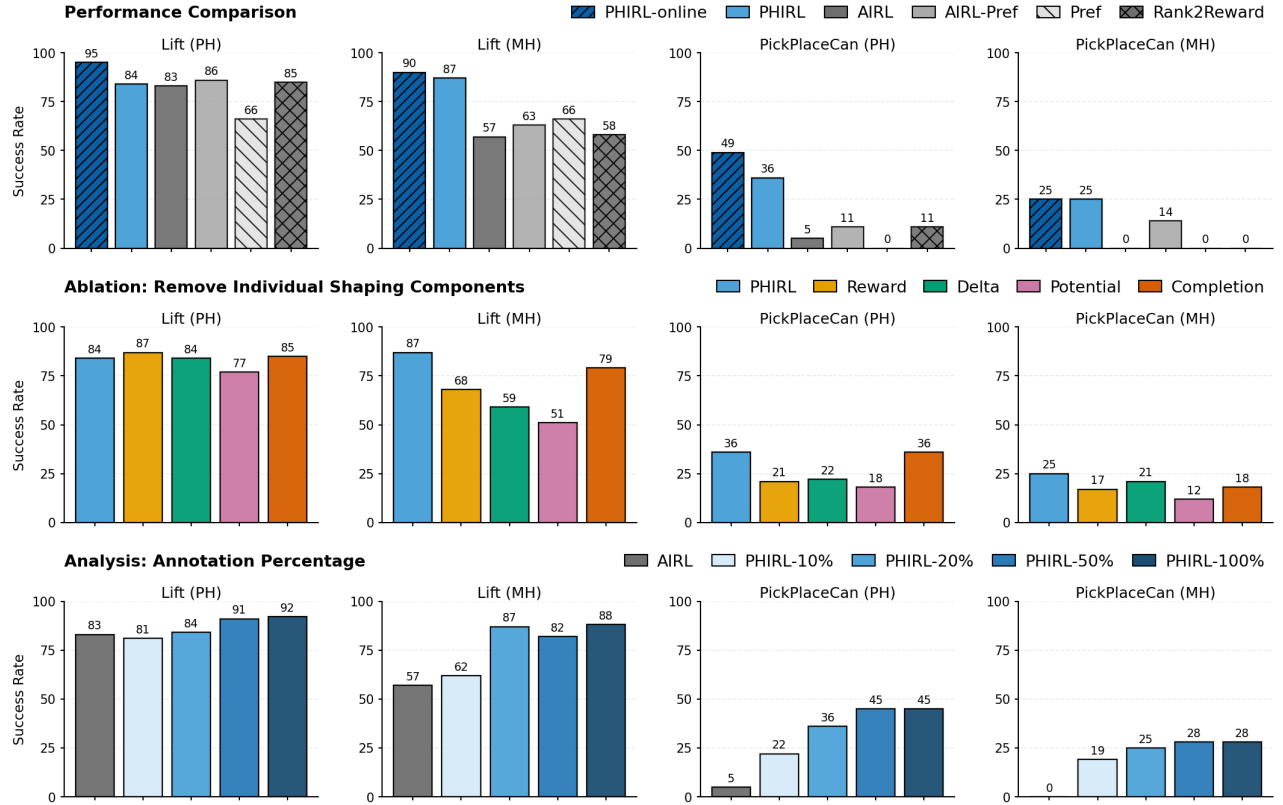


Fig. 1. Success Rates. The Y-axis indicates the success rate over 100 runs. Task success was decided by the environmental reward. When the agent has achieved success, the single-step return reward is 1.

in Figure 3. The results of reward learning experiments are presented in ??.

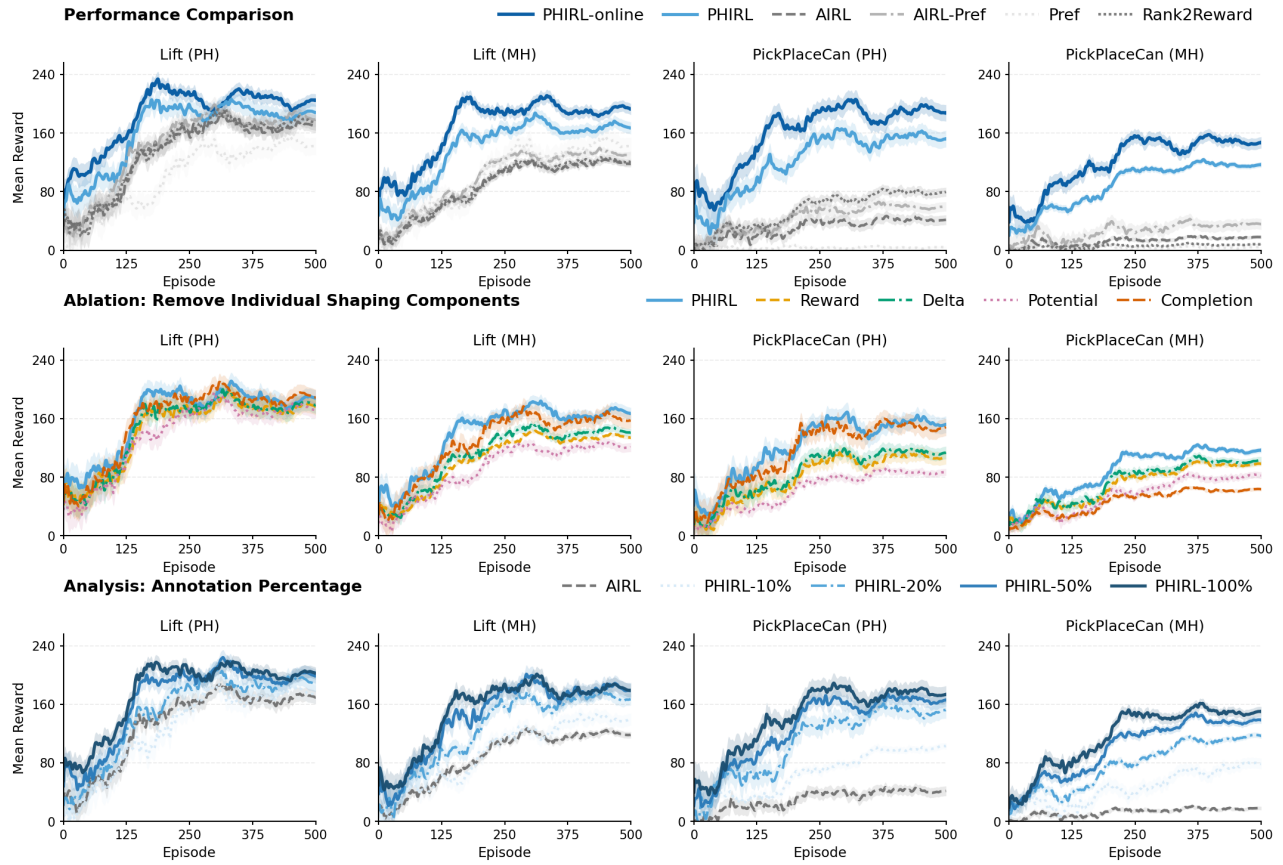


Fig. 2. Learning Curves. The Y-axis is the return rewards. The shaded area is the confidence interval.

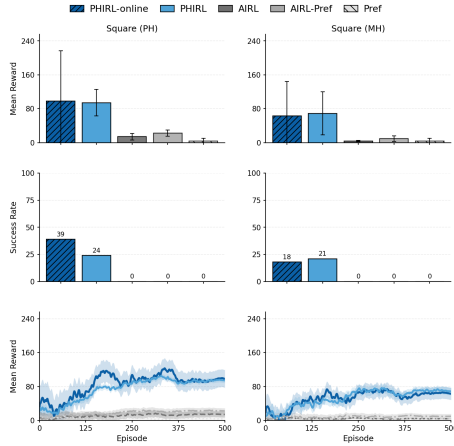


Fig. 3. Square Results. First row: average return rewards; second row: success rate over 100 runs; third row: reward curves. The error bars in row one are standard deviations. The shaded areas in row three are confidence intervals. PHIRL and PHIRL-online are the only methods that achieved task success.

The top row presents the average environmental return rewards of the last checkpoint across all baselines; the middle row details the success rate for each method; and the bottom row provides the learning curves for all the baselines. The columns, from left to right, correspond to the datasets: PH

and MH. Since Pref does not learn from demonstrations, we used the same data for PH and MH conditions. The shaded areas are confidence intervals. The error bars represent the standard deviation, and the reported rewards are smoothed by averaging over 100 evaluation rollouts executed with the best checkpoint using a set of seeds (1 to 100).

Across both datasets, despite the success rates are relatively low, PHIRL and PHIRL-online are the only methods that achieved task success. With the PH dataset, PHIRL-online significantly improved the task performance compared to PHIRL (39 success vs. 24 success), while there was no improvement with the MH dataset. This is contradictory to our previous observations (PHIRL-online would improve the performance more if the demonstration quality is low). We hypothesize that this is because most failures happened when the robot dropped the square onto the peg, and the overlap of the square and the peg creates out-of-distribution visual data, which leads the VLM to produce unreliable progress labels (PHIRL-online’s performance in both datasets shows a larger variance).

D. real robot study setup

In this task, the users were asked to use an Xbox controller to control a Kinova Gen 3 Lite robot arm (6 DoF) to brush all the pom-poms off the stuffed animal. An overhead camera and a scene camera are used for detecting the number of

pom-poms.

Demonstration Collection. We constructed our dataset using a similar structure as the Robomimic MH dataset [2]. We collected demonstrations from six participants, where two participants are the system developers (proficient users), two participants are people with computer science backgrounds (expert users), and two participants have no computer science background and no robot experience (non-expert users). Each of them provided 10 demonstrations. Before providing demonstrations, each participant has five minutes to practice.

Annotation Collection We follow the same protocol for collecting human annotations as in simulation experiments. The system randomly selected two out of ten demonstrations and asked participants to give *progress* annotations. PHIRL received 120 *progress* labels, Pref received preference feedback for 600 trajectory pairs, and AIRLPref received preference feedback for 120 trajectory pairs. PHIRL Online received an additional 100 online *progress* labels during the training from the VLM. The procedure and prompts used for fine-tuning the Gemma 12B model are the same.

REFERENCES

- [1] Y. Zhu, J. Wong, A. Mandlekar, R. Martín-Martín, A. Joshi, K. Lin, A. Maddukuri, S. Nasiriany, and Y. Zhu, “robosuite: A modular simulation framework and benchmark for robot learning,” 2025. [Online]. Available: <https://arxiv.org/abs/2009.12293>
- [2] A. Mandlekar, D. Xu, J. Wong, S. Nasiriany, C. Wang, R. Kulkarni, L. Fei-Fei, S. Savarese, Y. Zhu, and R. Martín-Martín, “What matters in learning from offline human demonstrations for robot manipulation,” 2021. [Online]. Available: <https://arxiv.org/abs/2108.03298>